# Pitambar Acharya

## Ph.D. Candidate in Applied Mathematics

*Department of Mathematics*
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Ph.D. candidate in Applied Mathematics with extensive undergraduate teaching experience in Precalculus, Calculus, Probability, Statistics, Linear Algebra, and foundational mathematics courses. Committed to student-centered teaching, fostering mathematical understanding, and creating an inclusive and engaging learning environment. Research expertise in nonlinear partial differential equations, mathematical modeling, computational mathematics, data science, and machine learning, with a strong interest in integrating real-world applications into undergraduate mathematics education.

## Research Interests

Partial Differential Equations • Data Science • Machine Learning • Scientific Computing

## Education

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|---|---|
| Aug 2022 – Expected Spring 2027 | **PhD in Applied Mathematics**<br>University of Alabama at Birmingham, Birmingham, AL<br>Advisor: Prof. Yanni Zeng<br>Minor: Computer Science |
| Aug 2025 | **Master of Science in Mathematics**<br>University of Alabama at Birmingham, Birmingham, AL |
| Jan 2020 – Jul 2022 | **Master's in Applied Mathematics and Statistics (Biostatistics Track)**<br>Florida Atlantic University, Boca Raton, FL<br>Advisor: Prof. Hongwei Long |
| 2012 – 2014 | **Master of Science in Pure Mathematics**<br>Tribhuvan University, Kirtipur, Kathmandu, Nepal |
| 2010 – 2012 | **Master of Science in Physics**<br>Tribhuvan University, Kirtipur, Kathmandu, Nepal<br>Advisor: Prof. Jeevan Nakarmi |
| 2007 – 2010 | **Bachelor of Science in Physics**<br>Trichandra College, Ghantaghar, Kathmandu, Nepal<br>Minor: Mathematics and Statistics |

## Teaching Experience

Aug 2022 – Present

**Instructor of Record**
University of Alabama at Birmingham, Birmingham, AL

- Precalculus: Algebra/Trigonometry (MA 107 QL) – Summer 2026
- Calculus I (MA 125/125L) – Spring 2026, Spring 2027
- Precalculus: Trigonometry (MA 106) – Fall 2026, Spring 2025, Fall 2024
- Introduction to Statistics (MA 180) – Fall 2025, Summer 2025
- Precalculus: Algebra (MA 105) – Spring 2024, Fall 2023, Spring 2023
- Intermediate Algebra (MA 102) – Fall 2022

Aug 2022 – Present

**Graduate Teaching Assistant**
University of Alabama at Birmingham, Birmingham, AL

- Tutored and proctored exams in the Math Learning Lab (MLL)

Jan – May 2022

**Instructor of Record**
Florida Atlantic University, Boca Raton, FL

- Methods of Calculus – Spring 2022

Jan 2020 – Jul 2022

**Graduate Teaching Assistant**
Florida Atlantic University, Boca Raton, FL

- Calculus II (MAC 2312) – Spring 2021, Fall 2020
- Calculus I (MAC 2311) – Spring 2020
- Engineering Mathematics I (MAP 3305) – Summer 2020
- Intermediate & College Algebra: quizzes and midterms – Fall 2021
- Tutor, Engineering Mathematics and Mathematics for Liberal Arts – Summer 2021
- Tutored and proctored exams in the Math Learning Center (MLC)
- Recitation Leader for Calculus II and III

Jun 2015 – Dec 2019

**Lecturer**
Himalaya College of Engineering (Tribhuvan University), Nepal

- Probability and Statistics, Statistic I & II
- Engineering Mathematics (covers Calculus I, II & III, Linear Algebra, Differential Equations)
- Engineering Physics (lectures and labs): Civil, Architecture, Computer, and Electronics Engineering students

Jun 2015 – Dec 2019

**Adjunct High School Teacher**
Kathmandu Model Higher Secondary School, Kathmandu, Nepal

- Basic Mathematics for Grade 11 & 12

## Publications

### Peer-Reviewed Journal Articles

[1] **P. Acharya**, and Y. Zeng: "Global Existence of Solutions on the Quarter Plane for a Chemotaxis Model with Logistic Growth and Logarithmic Sensitivity," *Manuscript in preparation*, University of Alabama at Birmingham.

[2] K. Acharya and **P. Acharya**. "Adaptive Filtering via Canonical Systems with Time-Varying Hamiltonians." *International Journal of Advances in Engineering Sciences and Applied Mathematics* (Springer Nature), 18, 20–30 (2026). DOI: 10.1007/s12572-026-00421-5

[3] E. P. Poudel, **P. Acharya**, J. Kafle, and S. Khadka. "On One-Dimensional Advection–Diffusion Equation with Variable Diffusivity." *Applied Mathematics: A Journal of Chinese Universities* (Springer Nature), 41, 181–193 (2026). DOI: 10.1007/s11766-026-5149-9

[4] D. Dulal, M. Gallagher, C. Navasca, **P. Acharya** and N. Skowronski. "Leveraging Advanced Machine Learning to Predict Turbulence Dynamics From Temperature Observations at an Experimental Prescribed Fire." *IEEE Access*, 14, 112481-112498 (2026) DOI:10.1109/ACCESS.2026.3715743.

[5] E. P. Poudel, **P. Acharya**, J. Kafle, and S. Khadka. "Modeling and Analysis of Two-Dimensional Contaminant Dispersion in Turbulent Atmospheric Flows." Submitted to *Electronic Journal of Differential Equations*.

[6] D. Bastola, **P. Acharya**, and D. Dulal. "Predicting Anemia Among Under-Five Children in Nepal Using Machine Learning and Deep Learning." Submitted to *Public Health Nutrition*. Preprint

[7] K. Acharya, **P. Acharya**, and D. Bastola. "Random Schrödinger Operators on Data Graphs: A Data Science Perspective on Robust Spectral Embedding." Manuscript in preparation.

## Additional Research and Projects

[1] **P. Acharya**, L. Zeng, and M. Islam. "Comparative Analysis of Data Mining Techniques for Customer Segmentation and Anomaly Detection." Graduate final project.

[2] M. Islam, H. Akther Samia, **P. Acharya**. "Bin Packing Problem with Machine Learning Techniques." View Project

[3] **P. Acharya**, O. Francis, and M. Islam. "Deep Q-Learning for Optimizing Bin Packing Problem: A Deep Reinforcement Learning Approach."

## Presentations

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| Aug 2025 | "Learning the Solution Operator of the Heat Equation."<br>Summer School on Scientific Machine Learning, Brin Mathematics Research Center, University of Maryland, College Park |
| Feb 2025 | "On One-Dimensional Advection–Diffusion Equation with Variable Diffusivity."<br>MAA–FTYCMA Joint Conference, Embry-Riddle Aeronautical University, Daytona Beach |
| May 2022 | "Bayesian Model Averaging: An Overview."<br>MS Presentation, Florida Atlantic University, Boca Raton, FL |

## Conferences & Workshops Attended

### Selected Conferences (recent)

- Fourth Annual Joint Alabama – Florida Conference on Differential Equations, Dynamical Systems & Applications (JAF DEDS) – FIU, May 2026
- 37th Annual UA System Applied Mathematics Meeting – UAH, Nov 2025
- 3rd UNCG Virtual PDE Conference – UNC Greensboro, Oct 2025
- Ohio River Analysis Meeting (ORAM 14) – University of Cincinnati, Mar 2025
- Southeastern Analysis Meeting (SEAM 41) – University of South Florida, Mar 2025
- Modeling and Simulations in Fluids – Brown University (ICERM), Sep 2024
- Joint Mathematics Meetings (JMM) – Boston, MA, Jan 2023

### Summer Schools & Workshops

- Summer School: Scientific Machine Learning – Brin MRC, University of Maryland, Aug 2025
- Texas Summer School in Mathematical Physics – Texas A&M University, Jun 2025
- Texas Summer School in Mathematical Physics – Texas State University, Jul 2023
- CIMPA Summer School: Data Visualization, Modeling and Mathematical Tools – Tribhuvan University, Nepal, May 2023 *(Organizing Committee Member)*

## Leadership & Service

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| Aug 2024 – Present | **Secretary** – SIAM Student Chapter, UAB |
| Aug 2022 – Present | **Executive Member** – Nepalese Students Association, UAB |
| Jul 2019 | **Deputy Leader** – International Mathematics Olympiad (IMO), Bath, United Kingdom |
| Jul 2017 | **Deputy Leader** – International Mathematics Olympiad (IMO), Rio de Janeiro, Brazil |
| Feb 2017 – Dec 2019 | **Vice President** – Mathematical Association of Nepal (MAN)<br>• Trained high school students for the National Mathematics Competition<br>• Conducted examinations and prepared students for IMO<br>• Organized national and international conferences |
| Aug 2020 – 2021 | **Vice President** – Nepalese Students Association, FAU |

## Community Outreach

- Alabama Statewide High School Mathematics Contest – Feb 2026
- Math Talent Search Grader and Proctor – Oct 2025
- Alabama Statewide Math Ciphering Contest – Apr 2025
- High School Science Exhibition Judge, Birmingham, AL – Jul 2023
- Kangaroo Mathematics Competition, UAB – Mar 2023
- Math Day Proctoring and Grading, FAU – Jan 2021

- High School Science Olympiad, FAU – Feb 2020
- Blood Donation Program, Kanti Children's Hospital, Kathmandu, Nepal
- Donated stationery and sports materials for cancer-affected children, Kanti Hospital, through Astitwo Foundation, Nepal

## Awards & Honors

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| Apr 2025 | James Robert Ward, Jr. Memorial Endowed Award ($1,500) for excellence in Graduate Studies in Differential Equations / Mathematical Biology – UAB |
| Mar 2025 | Travel Award ($563.61) – Ohio River Analysis Meeting, University of Cincinnati |
| Jun–Jul 2024 | Professional Development Stipend ($1,000) – Blazer Core Curriculum, UAB |
| Jul 2023 | Travel Award ($1,066) – Texas Summer School in Mathematical Physics, Texas State University |
| 2022 – Present | Graduate Scholarship / Graduate Teaching Assistantship – UAB |
| 2020 – 2022 | Graduate Scholarship / Graduate Teaching Assistantship – FAU |
| 2012 – 2014 | MS Fellowship, Central Department of Mathematics, Tribhuvan University |
| 2010 – 2012 | MS Fellowship, Central Department of Physics, Tribhuvan University |

## Technical Skills

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| **Teaching/Computer Technology** | Canvas, Pearson(MyMathLab/MyStatLab), Webassign, MyOpenMath, StatCrunch, Zoom, Latex, Word, Excel, PowerPoint, Mathematica |
| **Programming Languages** | Python, R, MATLAB, SQL, Minitab |
| **Python Libraries** | Pandas, NumPy, Matplotlib, SciPy, Scikit-learn, TensorFlow, Keras |
| **Deep Learning** | PyTorch (building and training neural networks) |
| **Tools & Platforms** | Git, Linux, Jupyter Notebook, AWS, GCP |
| **Statistical Software** | R (ggplot2), Minitab |

## Professional Memberships

- American Mathematical Society (AMS)
- Society for Industrial and Applied Mathematics (SIAM)
- Nepal Mathematical Society (NMS)
- Nepal Physical Society (NPS)
- Association of Nepalese Mathematicians in America (ANMA)

## Professional Development

- Data Science and Machine Learning: Making Data-Driven Decisions – MIT Institute for Data, Systems, and Society (MIT IDSS) - Certificate of Completion
- CTL LIFT Teaching Certificate – Center for Teaching and Learning, UAB
- Developing Great Presentations – UAB Learning and Development, May 2025
- Flex/Online Teaching Orientation – Centre for Online and Continuing Education (COCE), FAU
- Teaching Assistant Basics for Online Courses – COCE, FAU
- Inquiry-Based Learning Workshop – Department of Mathematical Sciences, FAU
- CRLA Certification Levels 1 & 2 – Mathematics Learning Center, FAU
- 14th Annual iLead Conference – Student Union Hall, FAU

## Languages

- English *(professional proficiency)*
- Nepali *(native)*
- Hindi *(proficient)*

## References

**Dr. Yanni Zeng**
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**Dr. Shannon Starr**
Associate Professor
University of Alabama at Birmingham
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